

Amal Shiyas

✉ amalshiyazabdulrahman@gmail.com ☎ 9037101027 📍 Kollam 🌐 Amal Shiyas

PROFILE

Mechanical Engineering graduate with hands-on experience in manufacturing, hydraulics, industrial operations, and engineering design. Currently working at Wipro Infrastructure Engineering in the Hydraulics division, gaining practical exposure to assembly operations, quality standards, production workflows, and hydraulic systems. Experienced with CAD tools and basic simulation, along with hands-on exposure to industrial processes through internships. A quick learner with a collaborative mindset, eager to contribute to efficient manufacturing and continuous improvement initiatives.

EDUCATION

Mechanical Engineering Amrita Vishwa Vidyapeetham CGPA- 6.73/10	10/2022 – 2026 Kollam, India
High School and Higher Secondary Stratford Public School 12th with 78.9% 10th with 89.8%	2022 Kollam, India

EXPERIENCE

Apprentice Engineer Wipro Infrastructure Engineering - Working in the Hydraulics division supporting assembly and manufacturing operations. - Assisting in the assembly, inspection, and testing of hydraulic components and systems. - Following industrial quality standards and safety procedures during production activities. - Collaborating with production teams to improve workflow efficiency and product quality. - Gaining practical exposure to hydraulic systems, engineering drawings, and manufacturing processes.	06/2026 – Present Chennai, India
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PROJECTS

Hand gesture control Wheelchair prototype

Built a gesture-controlled wheelchair prototype that translates hand movements into real-time directional control, enabling intuitive mobility without traditional input devices.

Real time Spatio-Temporal pressure mapping in contact mechanics

Working on developing a real-time spatio-temporal pressure-mapping system using an FSR-based sensor array for contact mechanics applications, involving the calibration of a multi-sensor grid with Arduino and Time-Division Multiplexing, the creation of a 3D-printed prototype, and the integration of hardware–software modules for preliminary heat-map visualization, with ongoing testing to analyze stress concentrations and enhance system accuracy for future applications like gait analysis and tactile sensing.

INTERNSHIPS

United Electrical Industries Limited. Intern - Gained hands-on exposure to industrial manufacturing processes, including machining, assembly, welding and electroplating, improving my understanding of real factory operations and product development. - Learned quality control and testing methods for electrical components, strengthening my skills in inspection, evaluation, and standards compliance.	04/2025 – 05/2025 Kollam, Kerala
Kerala Minerals Metals Ltd Intern - Gained hands-on experience in titanium dioxide manufacturing, from mineral separation to final product - Studied the chloride route process for TiO₂ production and synthetic rutile manufacturing - Observed industrial-scale mineral processing operations and quality control procedures	08/2024 – 09/2024 Kollam, India

Intern

- Applied mechanical engineering concepts in practical industrial settings at KEL (Kerala Electrical & Allied Engineering Co.Ltd )
- Collaborated with industry professionals on day-to-day operations and maintenance procedures
- Gained practical exposure to industrial electrical and mechanical systems

SKILLS

Technical Skills

- AutoCAD
- SolidWorks
- ANSYS
- MATLAB
- MS Excel
- Engineering Drawing Interpretation
- Manufacturing Processes
- Hydraulic Systems
- Quality Control
- Production Operations

Soft Skills

- Communication
- Team Collaboration
- Problem Solving
- Leadership
- Adaptability
- Time Management

CERTIFICATIONS

Introduction to Generative AI with Google Cloud

10/2023

Platform: Udacity, part of Accenture Collaboration: Google Cloud

- Mastered fundamental concepts of generative AI and machine learning through Google Cloud certification

Computational Fluid Mechanics - Airflow Around a Spoiler

07/2025

Coursera

- Understand cloud-simulation technology and why it is so powerful.
- How to set up an incompressible fluid flow simulation around a spoiler and post-process the results.

EXTRACURRICULARS

IKR-Indian Kart Racing 2025

Participated in IKR 2025 (Indian Kart Racing) held at Coastt Speedway under ISIE – Imperial Society of Innovative Engineers, as part of Team Asphalt Racing. Contributed to design, fabrication, and racing activities, with the team securing awards in the Lightest Weight Category and achieving an All-India Rank of 13 among participating teams.

Executive Member

Kollam, India

ASME AVV

Executive Member, ASME (American Society of Mechanical Engineers)

Actively contributed to organizing technical events, workshops, and competitions aimed at enhancing technical knowledge and practical skills among students.

Collaborated with peers to foster a culture of innovation and engineering excellence.

Football

Active member of school football team (middle school and high school). Participated in district-level tournaments. Won multiple CBSE Sahodaya Football Championships

Track and Field

Excelled in track and field throughout school, specializing in long jump, 200m, and 400m events

Music

Achieved bronze medal/third place in light music competition at the prestigious CBSE Sahodaya Youth Festival during final school year

Photography and Editing

I enjoy taking photos and editing videos because it helps me improve my attention to detail and creativity. It also allows me to tell stories through visuals in a clear and effective way.

VOLUNTEERING

Experience as a Scribe

Served as a scribe for a student in my second year of college, assisting them during their university exams. This role enhanced my interpersonal communication skills and taught me how to effectively and sensitively communicate with others, while further nurturing my commitment to service.

Community Service

A Student Social Responsibilities initiative

Workshop Volunteer – Health and Wellbeing Lourde Mata School, Chavara

- Conducted an engaging workshop for school students on health and wellbeing.
- Helped raise awareness about healthy habits and lifestyle choices among young learners.
- Developed communication and presentation skills through interactive sessions.

Amritavarsham 2025

Volunteered at Amritavarsham celebrations in Amritapuri, supporting event coordination, crowd management, and logistics, ensuring smooth operations during large-scale cultural and spiritual programs. Gained hands-on experience in teamwork, communication, and handling high-responsibility tasks in a fast-paced environment.

LANGUAGES

English

Malayalam